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NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations****Supplementary Examinations - December 2025****21BT504 / 20BT602 - UPSTREAM PROCESSING TECHNOLOGY**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the contribution of rDNA technology in improvement of plant and animal cells.	10	L*2	1	2,3
	b) Outline the methods in isolation and screening of microbial cells.	10	L2	1	2,3
2.	a) Discuss the techniques for the improvement of microbial cells through selection of mutants.	10	L2	1	2,3
	b) Explain the plant cells improvement through somatic hybridization.	10	L2	1	2,3
Unit – II					
3.	a) Enumerate the media formulation parameters for plant cell culture.	10	L2	2	2,3
	b) Describe the sterilization methods employed in plant and microbial cell cultures.	10	L2	3	2
4.	a) Discuss response surface methodology in process optimization with relevant design examples.	10	L2	2	2,3,5
	b) Highlight the importance of filter sterilization along with its types and selection criteria.	10	L2	3	2
5.	a) Appraise the design of batch sterilization and Del factor calculation.	10	L2	3	2
	b) Explain the media optimization principles with suitable designs.	10	L2	2	2,3,5
Unit – III					
6.	a) Outline the principles in aseptic inoculation to pilot plant fermenter.	10	L2	5	2
	b) Discuss the various animal cell culture processes.	10	L2	5	2
7.	a) Explain the steps in inoculum development and scale-up.	10	L2	4	2
	b) Describe the scale-up considerations in plant cell/tissue cultures.	10	L2	4	2
8.	a) Explain the mycelial processes for various industrial applications.	10	L2	5	2
	b) Describe the Leudeking – Piret models to describe the biomass kinetics.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
